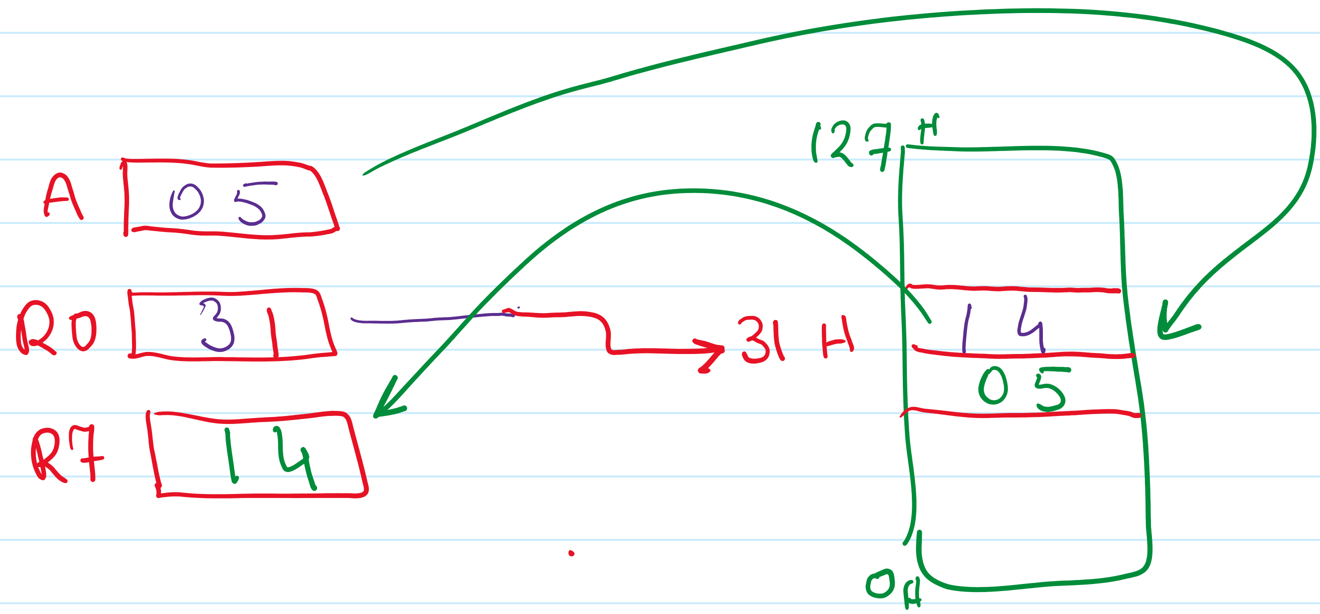
w-6

Tuesday, October 17, 2023 8:40 AM




```
MOV A, #5
MOV R7, #40H
MOV R0, #30H
MOV 31H, #14H
MOV @RO, A
INC R0
MOV R7, @R0
```

$$f = 12\ \text{MHz}$$

$$T = \frac{1}{12 \times 10^{6}}$$

$$= \frac{1}{12} \times 10^{-6}$$

$$T = \frac{1}{12}\ \mu s$$

$$24 \times T = 2\ \mu s$$

```
         MOV R0, #0        ;12 clocks
         MOV R1, #200      ;12 clocks
LOOP:    DJNZ R0, LOOP     ;256 * 24 clocks
         DJNZ R1, LOOP     ;executes inner loop + DJNZ 200 times
```

- Execution time is (12 + 12 + 200((256*24) + 24))/12000000 = 0.102802 sec
  0.000010
- Rewrite the code to generate a delay of 0.1usec accurate to 10usec

DJNZ Exercise

```
         MOV R0, #0
         MOV R1, #0
         MOV R2, #10
LOOP:    DJNZ R0, LOOP
         DJNZ R1, LOOP
         DJNZ R2, LOOP
```

1. How long does the above code take to execute if the 8051 is operating off a 12MHz crystal?
2. Repeat part 1 for a 16MHz crystal
3. Rewrite the code to generate a delay of 1 second accurate to 10usec (assume a 12MHz crystal)

05 18 22
08 12 21
07 16
06
02

02